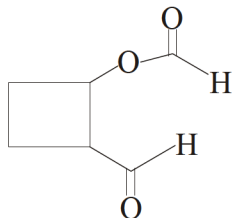


WORK SHEET - 02

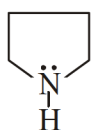
HINTS & SOLUTIONS

Q.1 (C)



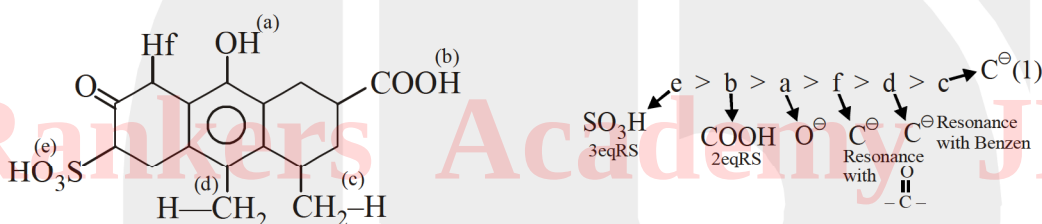
Correct IUPAC name: 2-Formylcyclobutyl methanoate

Q.2 (C)

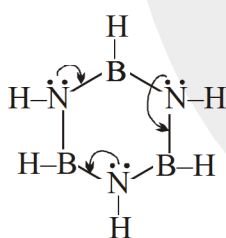


lone pair localised and present on sp^3 hybridised N-atom.

Q.3 (D)



Q.4 (C)



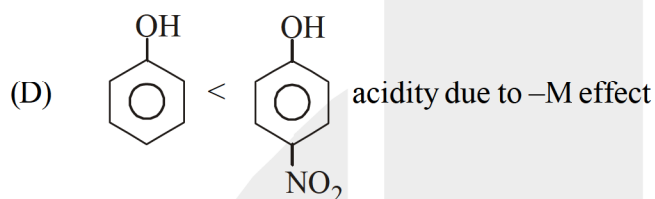
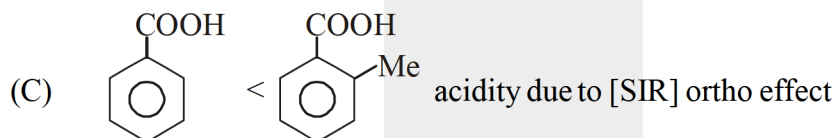
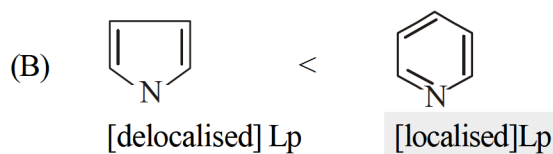
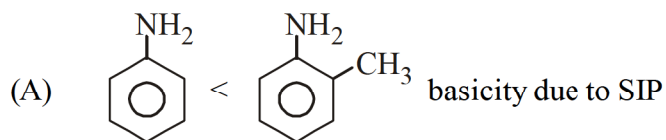
Results in 6π ($4n + 2$) π electron in a cyclic conjugated system (Aromatic)

Q.5 (C) (D)

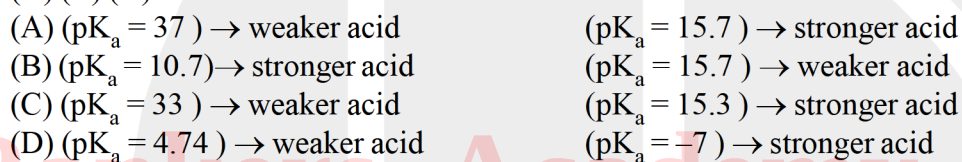
Q.6 (C)

Q.7 (C)

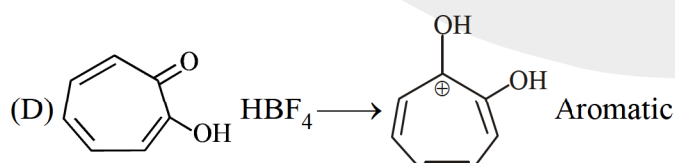
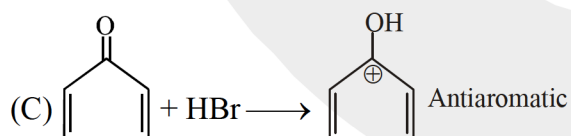
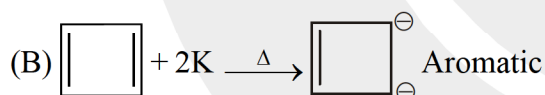
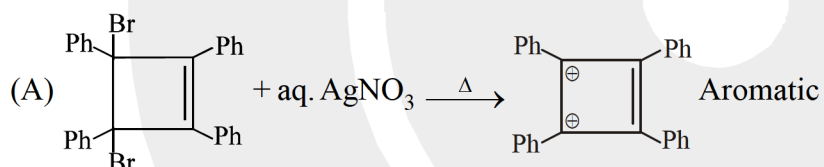
Q.8 (B) (C) (D)



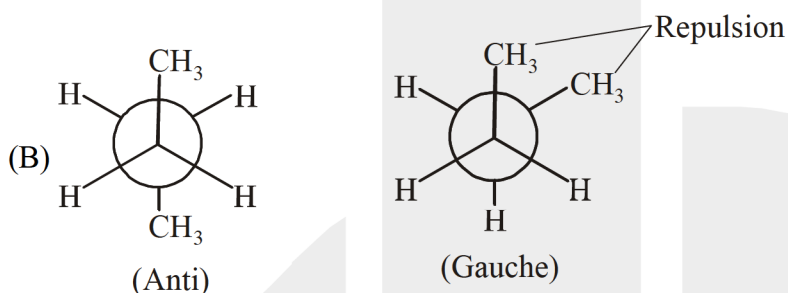
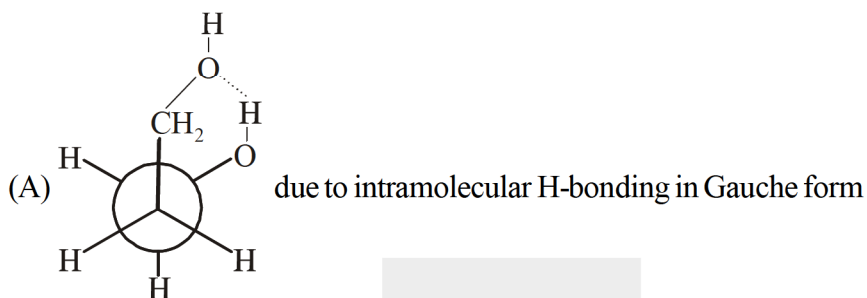
Q.9 (A) (C) (D)



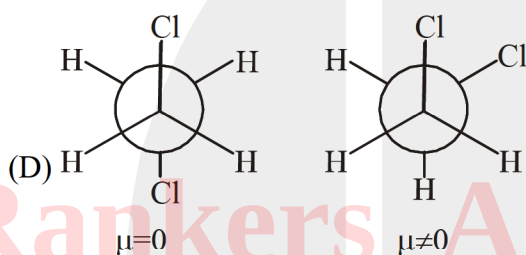
Q.10 (A) (B) (D)



Q.11 (A) (B) (C)



(C) Stability of eclipsed form is less than staggered at room temperature.

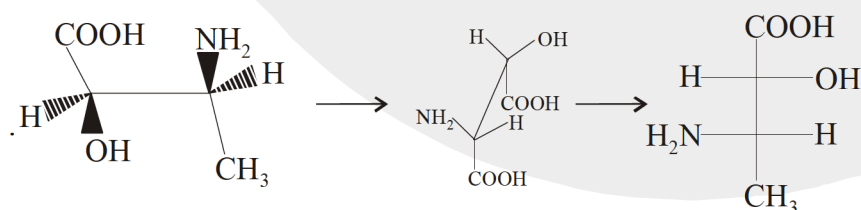


On heating rate of interconversion from anti-Gauche increases therefore μ increases.

Q.12 (A) (B) (C) (D)



Q.13 (A) P (B) R (C) Q



Q.14 4282

Given ΔH_H of $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH=CH}_2$ is 30 Kcal

(a) ΔH_H expected of $\text{CH}_2=\text{CH--CH=CH--CH}_3$ is 60 Kcal

But ΔH_{obs} of 1,3-pentene diene is 56 Kcal

R.E. = $\Delta H_{\text{obs}} - \Delta H_{\text{cal}} = 4$ Kcal

(b) ΔH_H of penta-1,3-diene is 56 Kcal

hence ΔH_H per π bond = 28 Kcal

(c) D.B.E = 2

Q.15 1321

(a) Aromatic

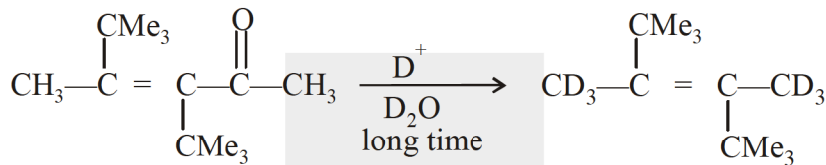
(b) Anti-aromatic

(c) Non-Aromatic

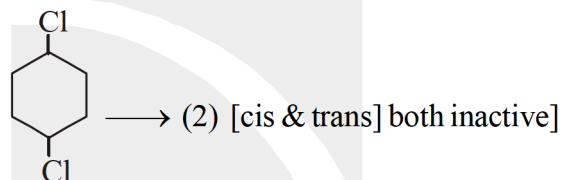
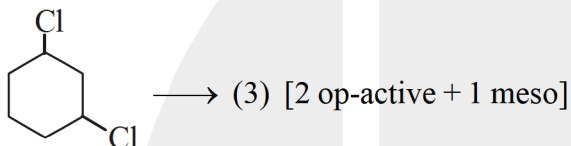
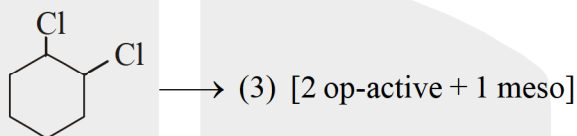
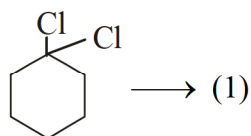
(d) Aromatic

Q.16 6

all H-atoms which are present in conjugation with $\text{—}\overset{\text{O}}{\parallel}{\text{C}}\text{—}$ are replaced by D.



Q.17 9



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